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SQL Fundamentals : SELECT & Filtering
Exercise 1 : SQL Foundations
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1. Retrieve all columns from the employee table :

Employees Table

id	First Name	Last Name	Department	Salary	Hire Date	City
1	Alice	Green	IT	70000	2020-01-10	Johannesburg
2	Brian	Lee	HR	45000		Cape Town
3	Cathy	Zulu	Finance	65000		Durban
4	David	Molcoena	Marketing	50000		Pretoria
5	Eva	Naidoo	IT	72000		Johannesburg

```
SELECT *
FROM employees;
```

2. Find all unique departments

```
SELECT DISTINCT department
FROM employees_table;
```

department
IT
HR
Finance
Marketing

3. Retriere first and last names ordered by salary descending

(2)

```
SELECT first_name, last_name
FROM employees
ORDER BY salary DESC;
```

first_name	last_nam
Eva	Naidoo
Alice	Green
Cathy	Zulu
David	Molotona
Brian	Lee

4. Retrieve the top 3 highest paid employees

```
SELECT id, first_name, last_name, salary
FROM employees
ORDER BY salary DESC
LIMIT 3;
```

id	first_name	last_name	salary
5	Eva	Naidoo	72000
1	Alice	Green	70000
3	Cathy	Zulu	65000

2. FILTERING with WHERE

5. Find employees in the IT department.

```
SELECT id, first_name, last_name, department
FROM employees
WHERE department = 'IT';
```

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id	first_name	last_name	department
1	Alice	Green	IT
5	Eva	Naidoo	IT

6. Find employees in Finance with salary > 6000

```
SELECT id, first_name, last_name, department, salary
FROM employees
WHERE department = 'Finance'
 AND salary > 60000;
```

id	first_nam	last_name	department	salary
3	Cathy	Zulu	Finance	65000

7. Find employees in HR or Marketing

```
SELECT id, first_name, last_name, department
FROM employees
WHERE department = 'HR'
    OR department = 'Marketing';
```

id	first_name	last_name	department
2	Brian	Lee	HR
4	David	Mokoena	Marketing

8. Find employees not in IT

```
SELECT id, first_nam, last_name, department
FROM employees
```

WHERE NOT department = 'IT';

id	first_name	last_name	department
2	Brian	Lee	HR
3	Cathy	Zulu	Finance
4	David	Molcoena	Marketing

9. Find employees in IT, HR or Finance Using IN

```
SELECT id, first_name, last_name, department
FROM employees
WHERE department IN ('IT','HR','Finance');
```

id	first_name	last_name	department
1	Alice	Green	IT
2	Brian	Lee	HR
3	Cathy	Zulu	Finance
5	Eva	Naidoo	IT

10. Find employees in IT with salary > 65000 and city Johannesburg

```
SELECT id, first_name, last_name, department,
       salary, city
FROM employees
WHERE department = 'IT'
AND   salary > 65000
AND   city = 'Johannesburg';
```

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id	first_name	last_name	department	salary	city
1	Alice	Green	IT	70000	Johannesburg
5	Eva	Naudoo	IT	72000	Johannesburg